CHINESE-ENGLISH MEDICAL TERM TRANSLATION LLM STUDY

CLINICAL BERT TEAM

AI 380H - AI IN HEALTHCARE

UNIVERSITY OF TEXAS AT AUSTIN

AGENDA

01 INTRODUCTION

02 PURPOSE

03 OBJECTIVE

04 METHODOLOGY

05 DEMO/RESULT

06 CONCLUSION

INTRODUCTION

The dataset integrates the English terminology of Chinese medicine (internal draft) created by the People's Health Publishing House (PMPH), the WHO International Standard Terminologies on Traditional Medicine in the Western Pacific Region established by the World Health Organization (WHO), and the International Standard Chinese-English Basic Nomenclature of Chinese Medicine developed by the World Federation of Chinese Medicine Associations (WFCMS), with the objective of advancing the standardization of Chinese medicine terminology and facilitating international communication regarding Traditional Chinese Medicine (TCM). The dataset is **somewhat** cleaned, sorted, and merged using the Python pandas library and optical character recognition technologies. Finally, it's separated into 56 groups. A total of 16189 records were sorted and combined into 8975 terms.

This comprehensive effort aims to bridge the gap between Eastern and Western medical paradigms, enhancing understanding and collaboration in the field of healthcare. By standardizing terminology, the dataset serves as a critical resource for educators, practitioners, and researchers who strive to integrate Traditional Chinese Medicine with modern medical practices. Additionally, it supports the translation and interpretation processes, ensuring that the rich knowledge of TCM is accurately conveyed across linguistic and cultural boundaries. The careful categorization into 56 groups reflects the diverse and multifaceted nature of TCM, allowing for more targeted studies and applications. As the global interest in alternative and complementary medicine continues to grow, this dataset along with large language models will undoubtedly play a pivotal role in fostering innovation and expanding the accessibility of traditional healing wisdom.

PURPOSE

Solidify the complications of language barrier between english compared to chinese as determining an already existing tool could already have the capabilities of accomplishing this obstacle. This tool can then be used by medical teams (e.g. doctors, nurses, practitioners, etc.) to help non native patient understand said medical term.

OBJECTIVE

To compare modern large language models (llms) interoperability of chinese medical terms. In essence, we aim to understand from learning perspective of both native english speaker and native traditional chinese speaker acquiring knowledge of medical terms using the modern llms.

METHODOLOGY

ADDITIONAL CLEANING OF DATASET

CHOOSING LLM PROVIDER/ INDIVIDUAL LLMS

EMBEDDING MODEL FOR SIMILARITY SCORE

BUILDING CHATBOT

BENCHMARK ON F1-SCORE, PRECISION, AND RECALL

ADDITIONAL CLEANING OF DATASET

EnglishName_1	EnglishName_2	EnglishName_3
diagnosis	0	0
DIAGNOSTICS	0	0
0	name of an extra point	extra point of EX-CA
cold stroke	center cold (cold in the middle jiao)	cold in the middle
lily disease	lily disease	lily disease
...	...	...
Draft on an Inquiry into the Properties of Things	0	0
Treatise on Blood Patterns/Syndromes	0	0
Dialogue of Fisherman and Woodman about Medici...	0	0
Longevity and Life Preservation in Eastern	0	0
Spring Mirror Records	0	0

EnglishName_1	EnglishName_2	EnglishName_3
diagnosis		
DIAGNOSTICS		
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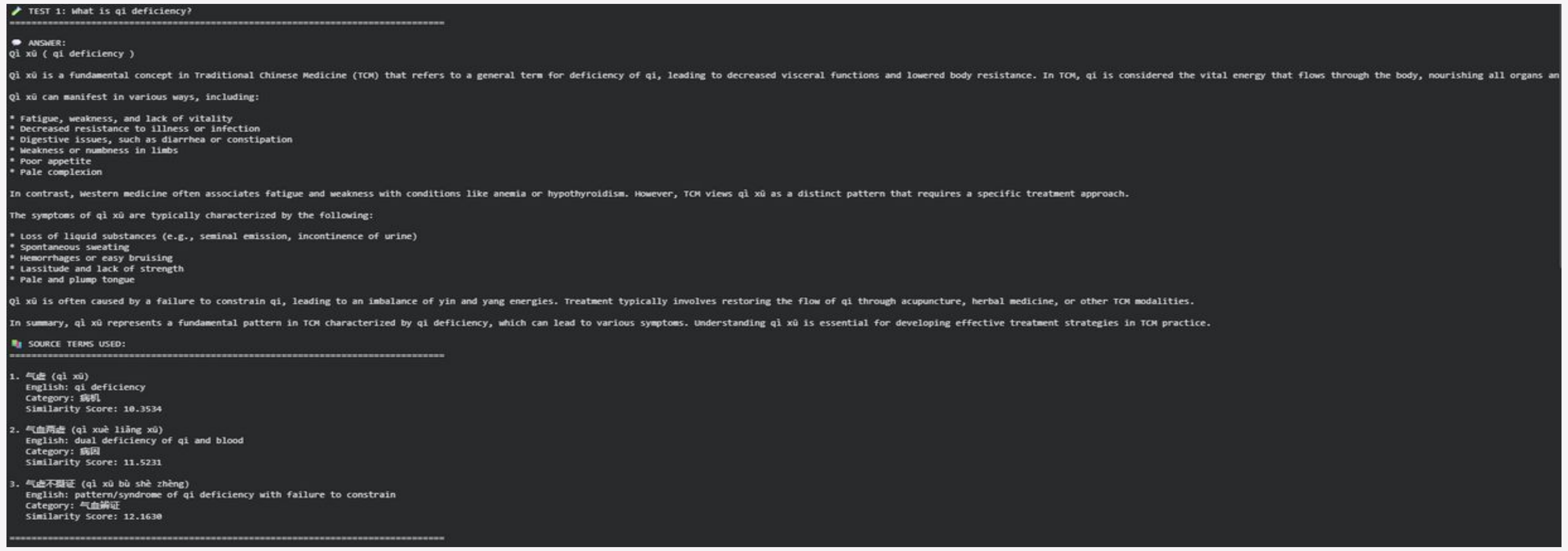
...	...	...
Draft on an Inquiry into the Properties of Things		
Treatise on Blood Patterns/Syndromes		
Dialogue of Fisherman and Woodman about Medici...		
Longevity and Life Preservation in Eastern		

Cleaned dataset: 8,971 terms with English translations (Removed 4 terms without English names) and we replaced 0s with empty string (“ ”)

CHOOSING LLM PROVIDER: OLLAMA

We harnessed the capabilities of Ollama, utilizing a suite of advanced models. Our approach incorporated to perform text generation given different queries related to the chinese medical terms to following open weight models: Deepseek-r1:8b, GPT-OS:20b Llama3.2:3b, Gemma-1:7b, and Yi:6b. This diverse selection allowed us to efficiently compare more well known (with exception of Yi:6b) large language models.



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TEST 1: What is qi deficiency?
================================================================================

ANSWER:
Qì xū ( qi deficiency )

Qì xū is a fundamental concept in Traditional Chinese Medicine (TCM) that refers to a general term for deficiency of qi, leading to decreased visceral functions and lowered body resistance. In TCM, qi is considered the vital energy that flows through the body, nourishing all organs an

Qì xū can manifest in various ways, including:

* Fatigue, weakness, and lack of vitality
* Decreased resistance to illness or infection
* Digestive issues, such as diarrhea or constipation
* Weakness or numbness in limbs
* Poor appetite
* Pale complexion

In contrast, Western medicine often associates fatigue and weakness with conditions like anemia or hypothyroidism. However, TCM views qì xū as a distinct pattern that requires a specific treatment approach.

The symptoms of qì xū are typically characterized by the following:

* Loss of liquid substances (e.g., seminal emission, incontinence of urine)
* Spontaneous sweating
* Hemorrhages or easy bruising
* Lassitude and lack of strength
* Pale and plump tongue

Qì xū is often caused by a failure to constrain qi, leading to an imbalance of yin and yang energies. Treatment typically involves restoring the flow of qi through acupuncture, herbal medicine, or other TCM modalities.

In summary, qì xū represents a fundamental pattern in TCM characterized by qi deficiency, which can lead to various symptoms. Understanding qì xū is essential for developing effective treatment strategies in TCM practice.

SOURCE TERMS USED:
================================================================================

1. 气虚 (qì xū)
   English: qi deficiency
   Category: 病机
   Similarity Score: 10.3534

2. 气血两虚 (qì xuè liǎng xū)
   English: dual deficiency of qi and blood
   Category: 病因
   Similarity Score: 11.5231

3. 气虚不摄证 (qì xū bù shè zhèng)
   English: pattern/syndrome of qi deficiency with failure to constrain
   Category: 气血辨证
   Similarity Score: 12.1630

================================================================================
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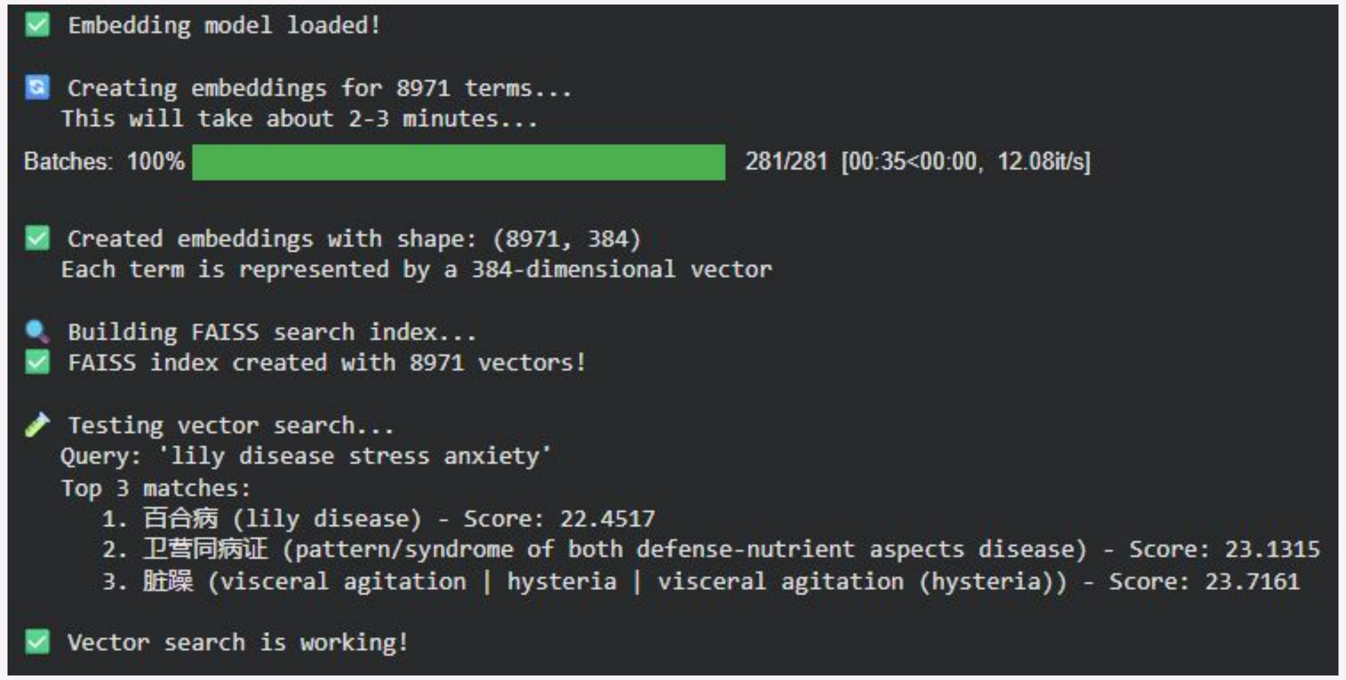
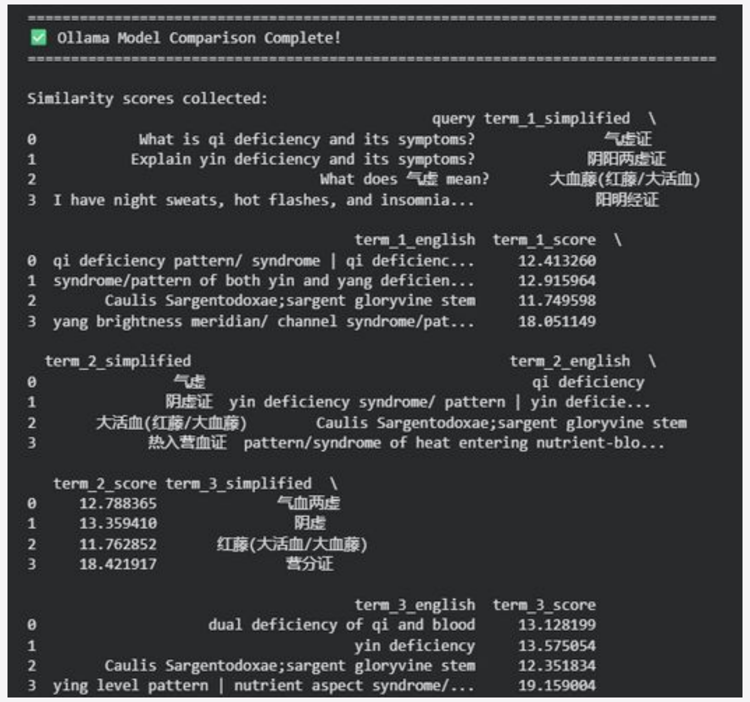
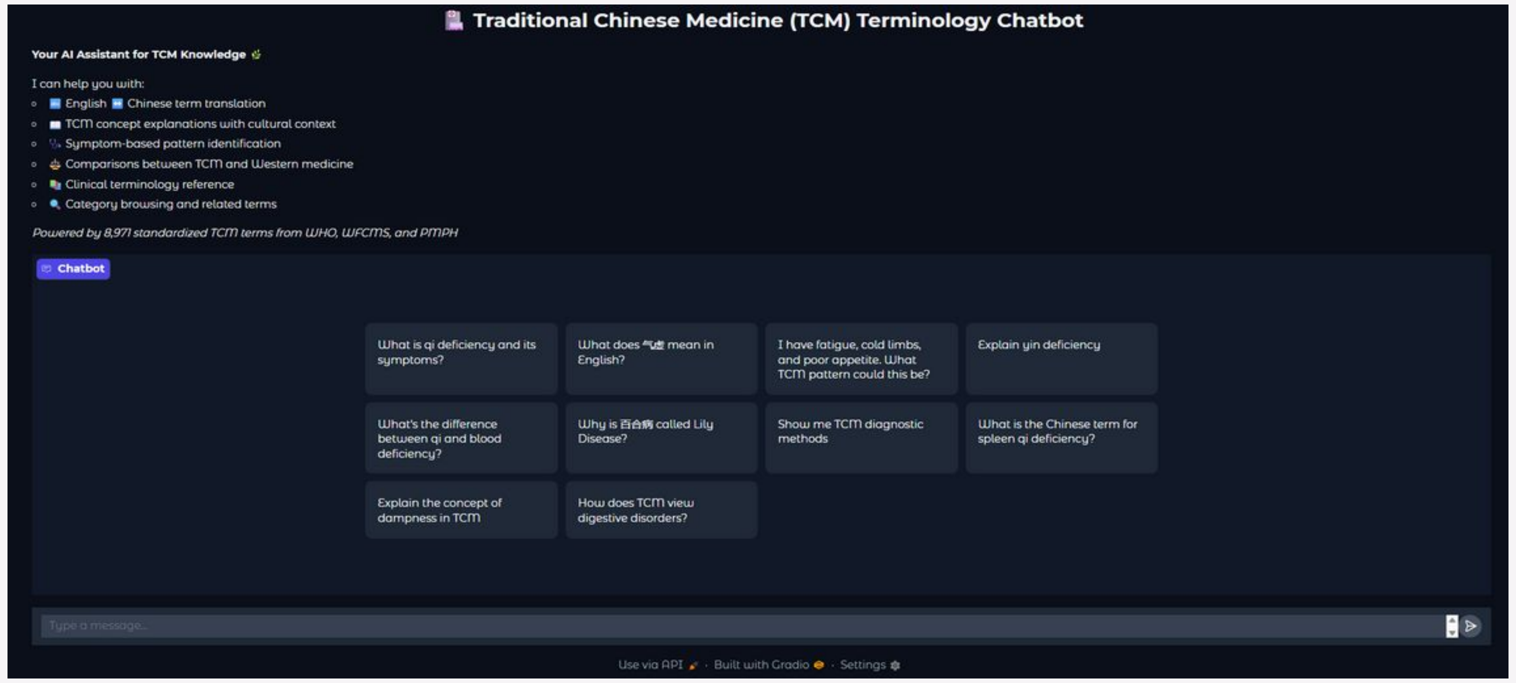
EMBEDDING MODEL FOR SIMILARITY SCORE





Our embedding model utilizes these terms to create vector representations that capture the semantic nuances of each word. By mapping these terms into a continuous vector space, the model efficiently computes similarity scores. These scores quantitatively reflect the relatedness of two terms within this semantic space, which supports tasks such as clustering, search optimization, and recommendation systems.

BUILDING CHATBOT



- We utilized the Gradio Python library to develop a chatbot powered by Ollama's open-weight models. Gradio's user-friendly interface enabled us to swiftly create an interactive web application, where users can engage with the chatbot in real-time. By integrating Ollama's models, which are designed for flexibility and efficiency, the chatbot is capable of understanding and responding to user input with remarkable accuracy.

- During the development process, we ensured the dataset was meticulously cleaned. This involved curating a list of 8,971 terms, each paired with English translations. We removed four terms that lacked English names to maintain consistency. Additionally, any placeholder zeros were replaced with empty strings to streamline data processing.

- The combination of Gradio's intuitive framework and Ollama's robust models resulted in a seamless and effective chatbot experience, demonstrating the potential of open-source tools in creating advanced conversational AI applications.

BENCHMARK ON F1-SCORE, PRECISION, AND RECALL

```
for model_name in ollama_models:
    response_text = model_responses[query_text].get(model_name, "")
    if not response_text:
        print(f"  No response found for {model_name} on query '{query_text}'.")
        continue

    results = evaluate_response_quality(response_text, ground_truth_for_query, flat_tcm_terms_list)

    tp = results['TP']
    fp = results['FP']
    fn = results['FN']

    precision = tp / (tp + fp) if (tp + fp) > 0 else 0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0
    f1_score = 2 * (precision * recall) / (precision + recall) if (precision + recall) > 0 else 0

    all_evaluation_results.append({
        'Query': query_text,
        'Model': model_name,
        'TP': tp,
        'FP': fp,
        'FN': fn,
        'Precision': precision,
        'Recall': recall,
        'F1-score': f1_score
    })
    print(f"  {model_name}: TP={tp}, FP={fp}, FN={fn}, P={precision:.2f}, R={recall:.2f}, F1={f1_score:.2f}")
```

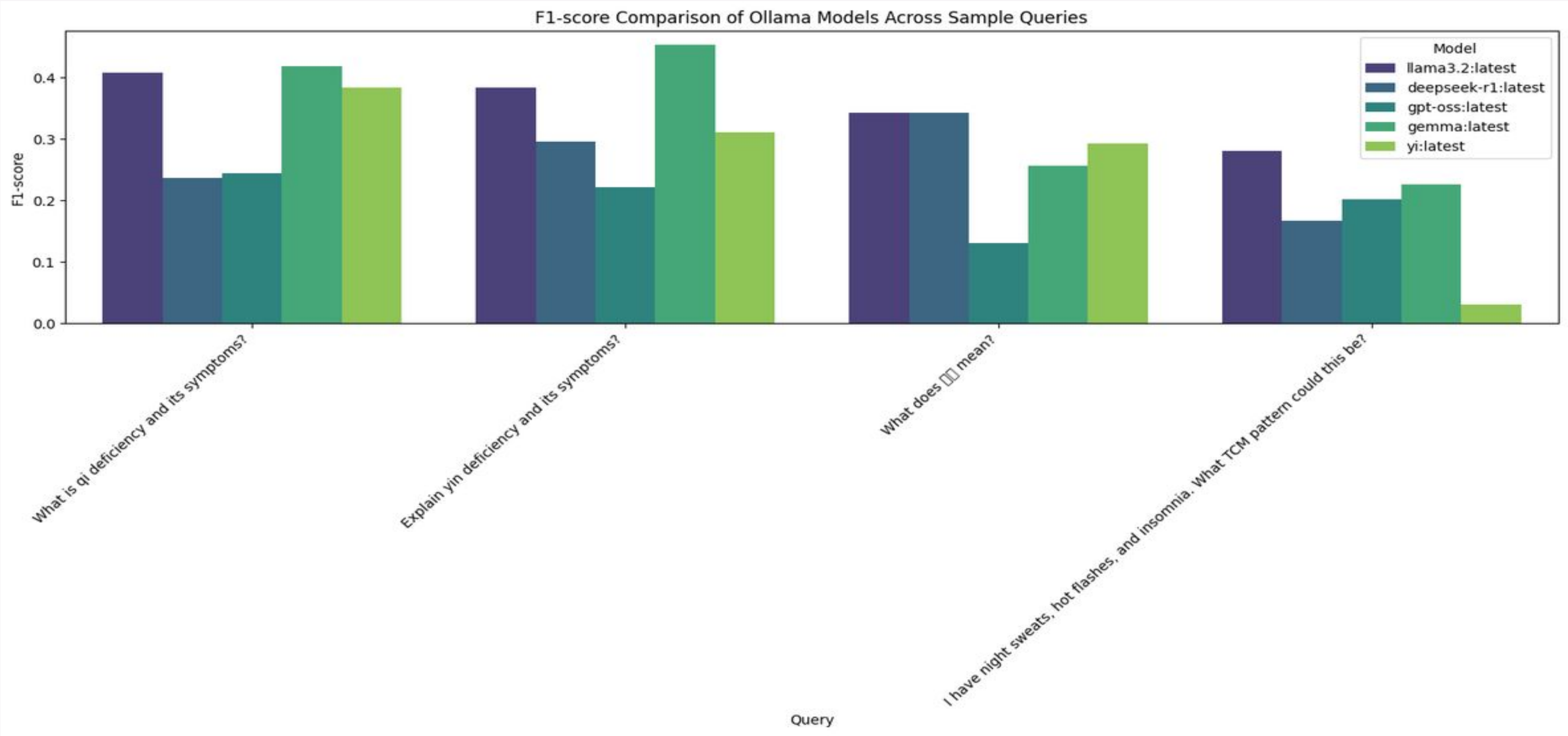
Using True Positives (TP), False Positives (FP) & False Negatives (FN) we were able to calculate F1-Score, Precision, and Recall

DEMO

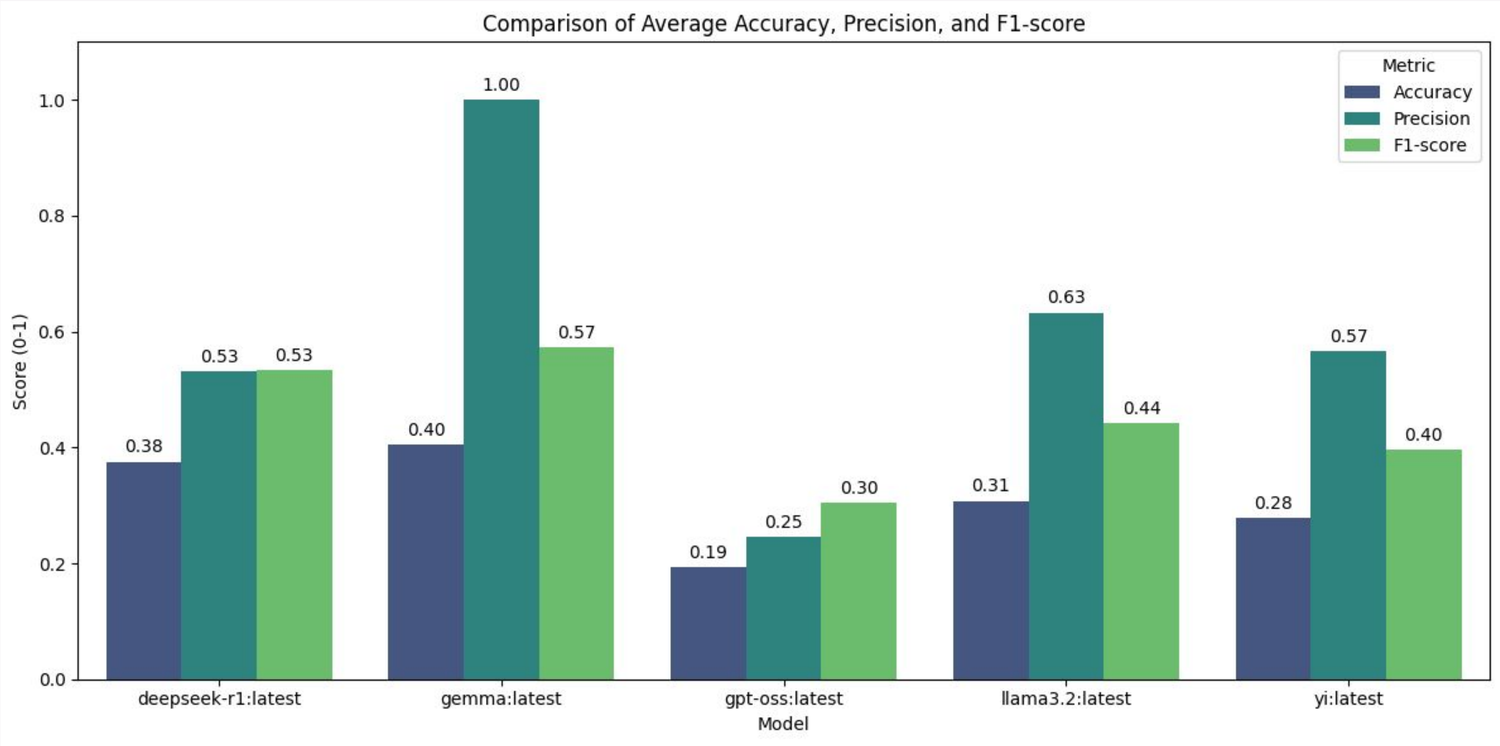
[AIHC.ipynb]

RESULT

- **MODELS COMPARED**: LLAMA3.2, DEEPSEEK-R1, GPT-OSS, GEMMA, YI
- **EVALUATION METRIC**: F1-SCORE ACROSS 5 TRADITIONAL CHINESE MEDICINE (TCM) QUERIES
- **KEY FINDINGS**:
  - **LLAMA3.2:1B** CONSISTENTLY ACHIEVED THE HIGHEST SCORES ACROSS MOST QUERIES
  - **GEMMA:7B AND YI:6B** SHOWED MODERATE PERFORMANCE WITH VARIABILITY DEPENDING ON QUERY TYPE
  - **DEEPSEEK-R1:8B AND GPT-0S:20B** GENERALLY SCORED LOWER, INDICATING WEAKER HANDLING OF TCM-SPECIFIC TERMINOLOGY
- **QUERY SENSITIVITY**: PERFORMANCE VARIED MOST ON SYMPTOM-BASED QUERIES (E.G., NIGHT SWEATS, HOT FLASHES, INSOMNIA)
- **OVERALL INSIGHT**: Gemma:7b DEMONSTRATED STRONGER DOMAIN UNDERSTANDING AND RELIABILITY FOR TCM-RELATED QUESTIONS



$$Precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{TP+FN}, \quad F1 = \frac{2(Precision \times Recall)}{Precision + Recall}$$



CONCLUSION

This study brought together authoritative sources of Traditional Chinese Medicine terminology, including PMPH, WHO, and WFCMS standards, to create a unified dataset. Through careful cleaning, sorting, and merging with Python and OCR technologies, 16,189 records were consolidated into 8,975 standardized terms organized into 56 thematic groups. This dataset provides a reliable foundation for evaluating AI models and advancing the international standardization of TCM terminology.

When tested across five representative queries, the models showed clear differences in performance. LLAMA3.2:1b consistently achieved the highest F1-scores, demonstrating stronger domain understanding and reliability in handling TCM-specific language. Gemma:7b and YI:6b performed moderately, with variability depending on the query type, while deepseek-r1:8b and GPT-oss:20b generally scored lower, indicating weaker handling of specialized terminology. Performance was most sensitive on symptom-based queries, such as those involving night sweats, hot flashes, and insomnia.

Overall, however, the average results highlight Gemma as the most effective model for TCM-related tasks given the current dataset. By combining a standardized terminology base with comparative model evaluation, this work supports both cross-cultural medical dialogue and the advancement of AI applications in healthcare.

THANK YOU